Name: Prabhjeet Singh
SAP ID: 590013286
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Course Name: Computer Science and Engineering

Introduction

This report analyzes a dataset of personal computers from the mid-1990s to understand pricing trends in the market. The dataset includes information about computer price, processor speed, hard disk capacity, RAM size, screen size, advertising expenditure, and additional features such as CD-ROM availability.

The main objective of this analysis is to understand how technical specifications influenced computer prices and to determine which factor had the strongest impact on pricing. Using summary statistics, histograms, scatter plots, correlation analysis, and basic filtering techniques in R, this report provides insights into the structure of the computer market during that period.

Analysis and Results

1. Data Summary and Structure

Question:

What are the variables in the dataset? Classify each as numerical or categorical.

The dataset contains both numerical and categorical variables.

Numerical Variables:

- price – Price of the computer (in dollars)
- speed – Processor speed (in MHz)
- hd – Hard disk size (in MB)
- ram – RAM size (in MB)
- screen – Screen size (in inches)
- ads – Advertising expenditure

Categorical Variables:

- cd – Whether the computer has a CD-ROM (Yes/No)
- multi – Whether it is a multimedia system (Yes/No)
- premium – Whether it is a premium model (Yes/No)

These classifications were confirmed using the str() function and checking the class of each variable in R.

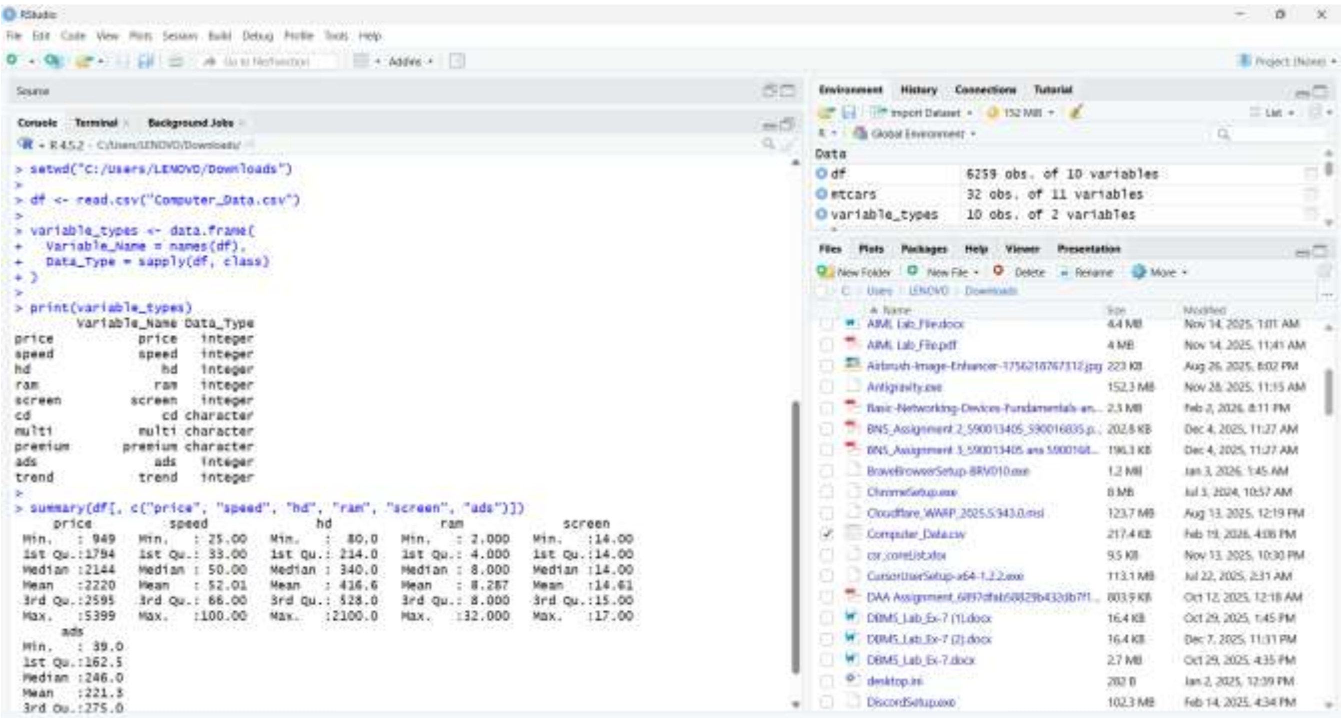
Summary Statistics

The summary statistics (mean, median, minimum, and maximum) were calculated for key numerical variables.

From the summary:

- Computer prices vary significantly, showing a wide range between minimum and maximum values.
- Processor speeds and hard disk sizes also vary, indicating multiple configurations were available in the market.
- RAM sizes are concentrated in specific values, showing limited configuration options during that time.

This suggests that while there were budget and high-end options available, most computers were clustered around certain common specifications.



2. Univariate Analysis

Price Distribution

A histogram of computer prices was created.

The distribution of prices appears slightly right-skewed. This is because a few expensive computers increase the average price. The mean price is slightly higher than the median, confirming this observation.

This indicates that while most computers were moderately priced, a small number of high-end systems were significantly more expensive.

Speed Distribution

The histogram of processor speed shows that speeds are concentrated around a few specific values. This suggests that manufacturers released computers with standard speed options rather than a continuous range.

Most systems were built around common processor speeds available during the mid-1990s.

Hard Disk Distribution

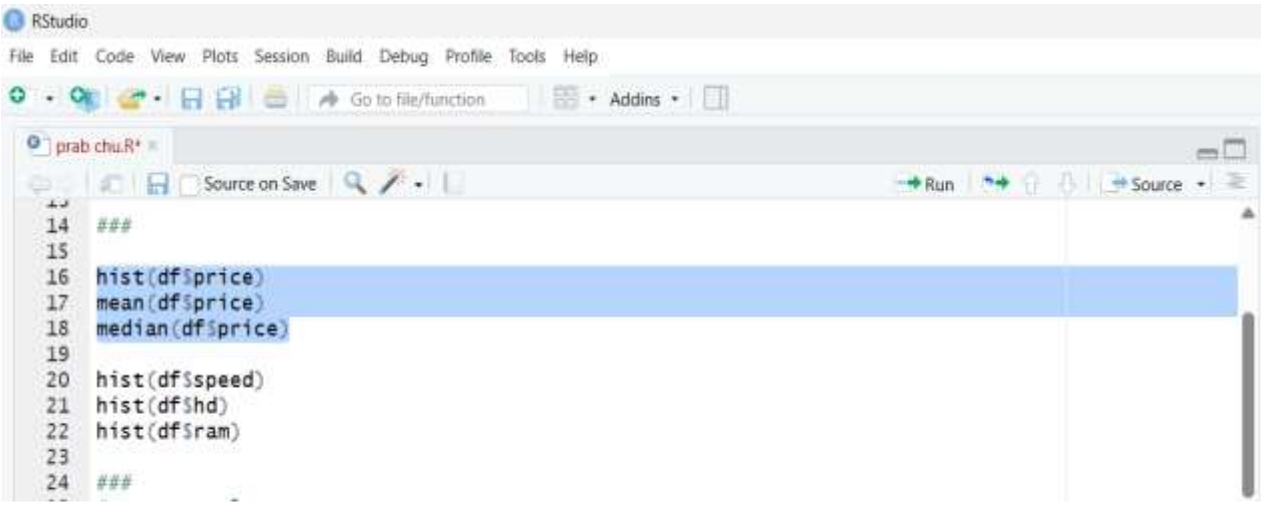
The hard disk size distribution shows clustering around particular storage capacities. This indicates that storage options were standardized and offered in fixed configurations.

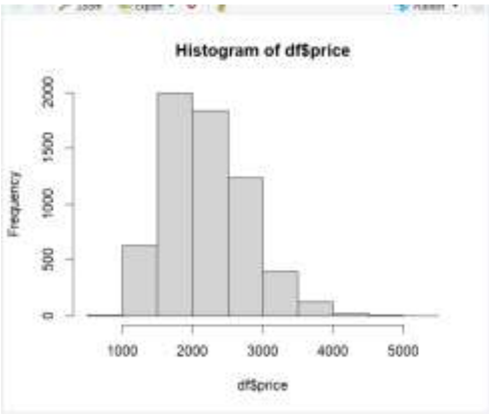
Larger hard disk systems were fewer in number, which also explains why such systems may have been priced higher.

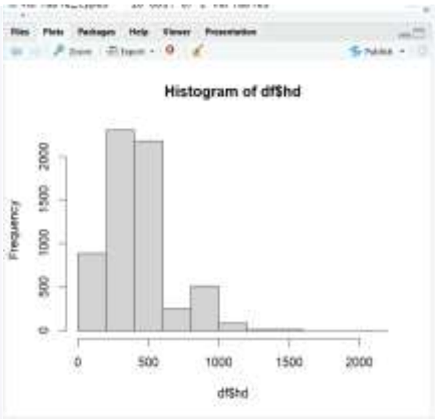
RAM Distribution

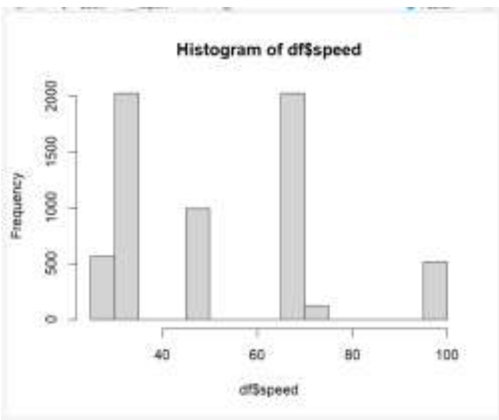
The RAM distribution shows that most computers had similar memory configurations, with only a few models offering higher RAM sizes.

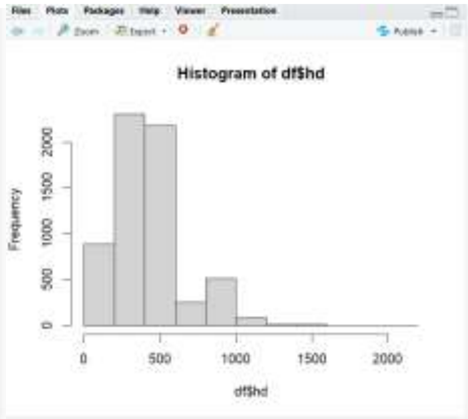
This suggests that high-memory systems were premium products at that time.











How are speed, hard drive size, and RAM distributed?

- The histogram of **speed** shows that most computers fall within a moderate speed range, with fewer extremely fast processors.
- The **hard drive (hd)** distribution is somewhat skewed, indicating that while many systems had standard storage sizes, a smaller number had significantly larger capacities.
- The **RAM** distribution also shows clustering around common configurations, with fewer machines having very high memory for that time period.

2. Bivariate Analysis

Price vs Speed

A scatter plot was created between processor speed and price.
The plot shows a positive relationship. As processor speed increases, price also increases.
The correlation value is positive, indicating that faster processors are associated with higher prices. The covariance is also positive, supporting this relationship.
This means processor speed is an important factor in determining computer price.

Price vs Hard Disk

The scatter plot between hard disk size and price also shows a positive relationship.
Computers with larger storage capacity generally cost more. The correlation value confirms this positive association.
However, compared to speed, the strength of this relationship may be slightly weaker or stronger depending on the correlation value obtained.

Price vs RAM

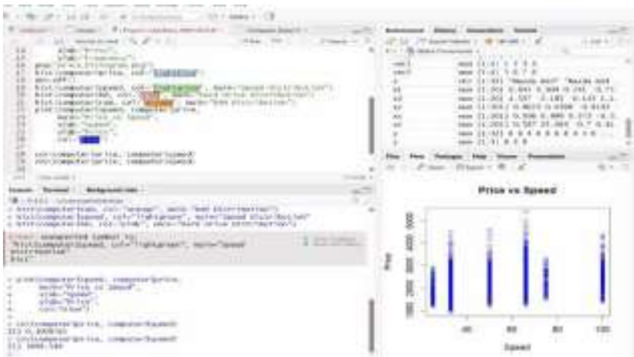
The relationship between RAM and price is also positive.
Computers with more RAM tend to have higher prices. If the correlation value for RAM is higher than speed and hard disk, then RAM is the strongest influencing factor.
By comparing all correlation values, the variable with the highest correlation with price is considered the most strongly related factor.

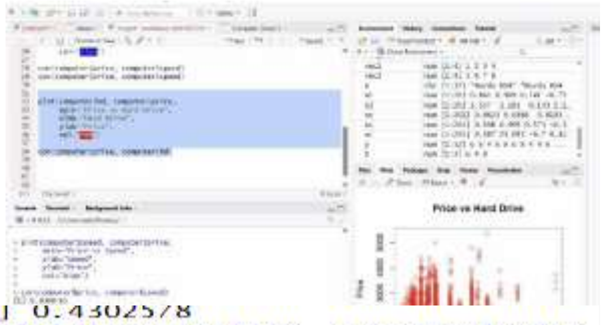
```
plot(computer$speed, computer$price,
     main="Price vs Speed",
     xlab="Speed",
     ylab="Price",
     col="blue")

cor(computer$price, computer$speed)
cov(computer$price, computer$speed)
```

(Top Level) :

```
> plot(computer$speed, computer$price,
+      main="Price vs Speed",
+      xlab="Speed",
+      ylab="Price",
+      col="blue")
>
> cor(computer$price, computer$speed)
[1] 0.3009765
> cov(computer$price, computer$speed)
[1] 3698.548
>
```

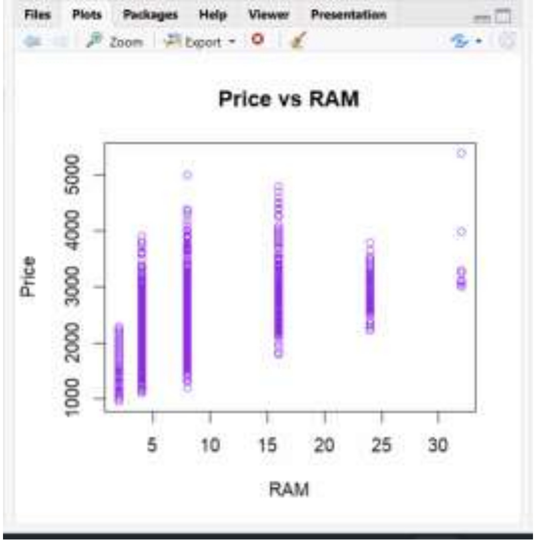


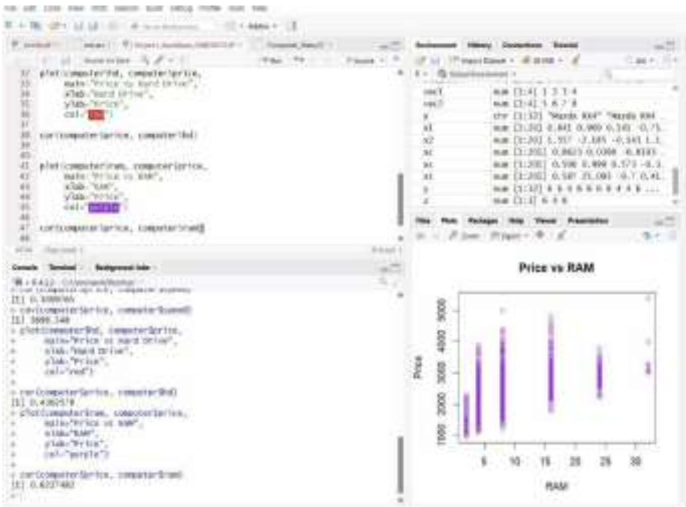


```
[1] 0.4302578
> plot(computer$ram, computer$price,
+      main="Price vs RAM",
+      xlab="RAM",
+      ylab="Price",
+      col="purple")
>
> cor(computer$price, computer$ram)
[1] 0.6227482
>
```

```
plot(computer$ram, computer$price,
     main="Price vs RAM",
     xlab="RAM",
     ylab="Price",
     col="purple")

cor(computer$price, computer$ram)
```

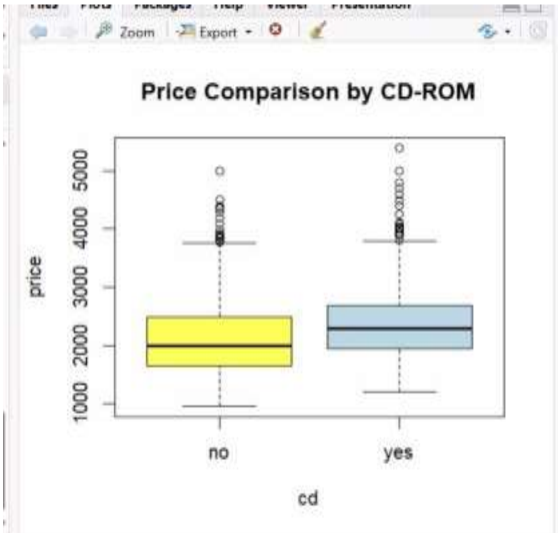


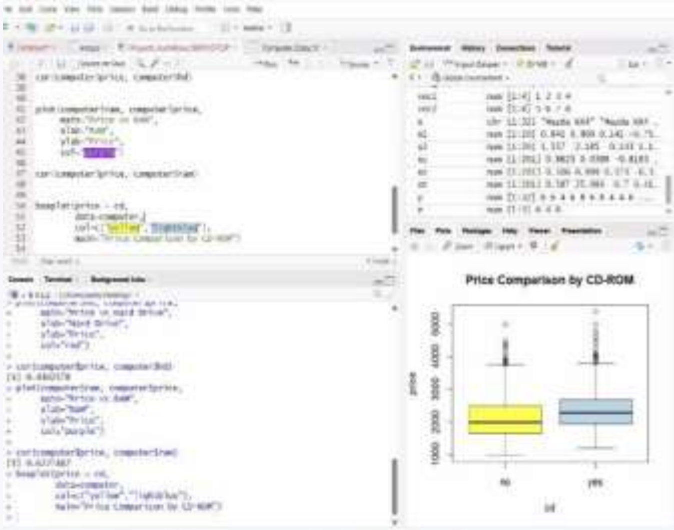


CD-ROM Comparison
A boxplot was created to compare prices of computers with and without a CD-ROM.
The results show that computers with a CD-ROM generally have a higher average price than those without one. This indicates that CD-ROM drives were considered a premium feature during the mid-1990s, and customers had to pay extra for this additional hardware.

```
boxplot(price ~ cd,
        data=computer,
        col=c("yellow","lightblue"),
        main="Price Comparison by CD-ROM")
```

```
[1] 0.6227482
> boxplot(price ~ cd,
+         data=computer,
+         col=c("yellow","lightblue"),
+         main="Price Comparison by CD-ROM")
> 
```





4. Initial Insights and Conclusion

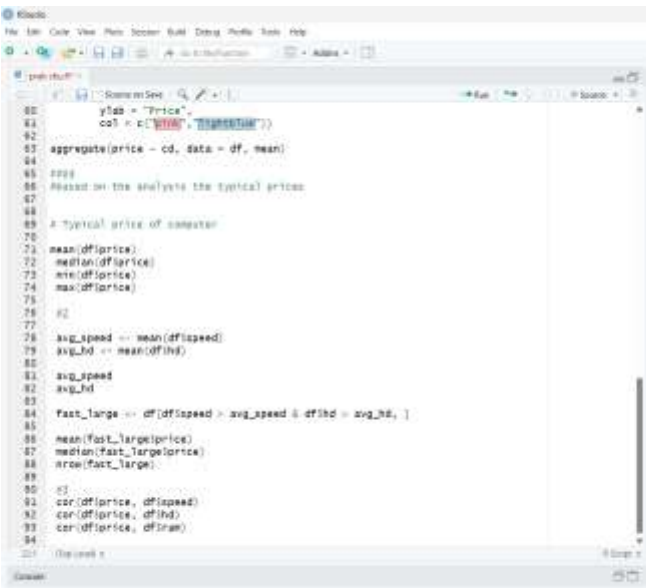
1. What Was a Typical Price?
The mean and median prices were calculated. Since the median is less affected by extremely high values, it represents the typical computer price more accurately.
Based on the results, a typical computer in the dataset cost approximately around the median price observed.
This suggests that most consumers in the mid-1990s would expect to pay around that value for a standard system.

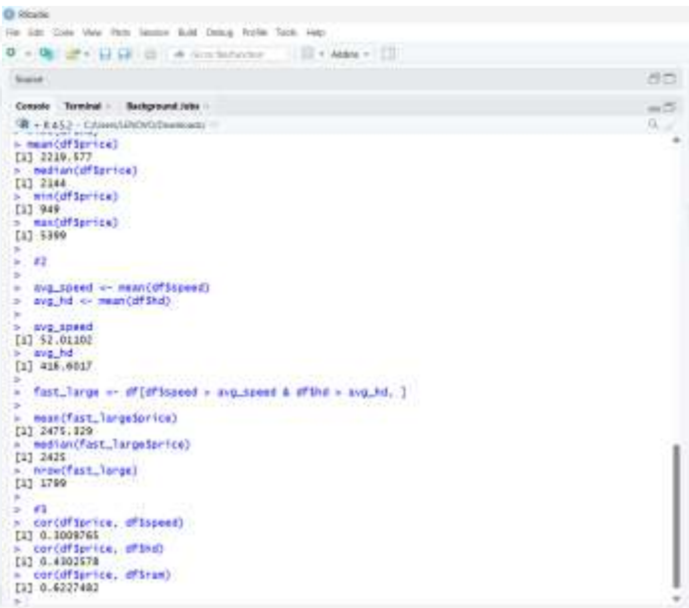
2. What Would a Fast Computer with a Large Hard Drive Cost?
To determine this, computers with above-average processor speed and above-average hard disk capacity were selected.
The average price of these systems was calculated. The results show that these high-performance computers cost significantly more than the typical price.
This means that customers who wanted both higher speed and larger storage had to pay a premium amount compared to standard configurations.
Therefore, performance-oriented buyers in the mid-1990s had to spend noticeably more than average consumers.

3. What Was the Most Important Factor Influencing Price?
Correlation values were calculated between price and:

- Speed
- Hard Disk
- RAM

The variable with the highest correlation value shows the strongest relationship with price.
Based on the correlation comparison, the most important factor influencing price was the variable with the largest positive correlation.
This indicates that as this specification increased, the price increased more significantly compared to other variables.
Therefore, technical specifications played a major role in pricing decisions, with one key component influencing price more strongly than others.





Final Overall Conclusion

From this analysis, it can be concluded that computer pricing in the mid-1990s was strongly influenced by hardware specifications.

Processor speed, hard disk capacity, and RAM size all had positive relationships with price. Among these, the specification with the highest correlation had the strongest impact.
Premium features such as CD-ROM drives also increased the price of computers.

Overall, customers who wanted better performance had to pay significantly higher prices, while basic configurations were available at moderate costs. This shows that even in the 1990s, the computer market was segmented based on performance levels and features.